

EXPERIMENT NO: 22

SERIAL TRANSMISSION AND RECEPTION USING ON-CHIP UART IN LPC2148 KIT

AIM

To write and execute C program for serial transmission and reception using on-chip UART in LPC2148 kit.

OBJECTIVES

- Initialize the on-chip UART0 of the LPC2148 for serial communication.
- Transmit a received character through the UART serial port.
- Verify serial transmission and reception using the UART data registers and status flags.

APPARATUS / SOFTWARE REQUIRED

- Keil uVision5 Software
- Philips Flash Programmer
- LPC 2148 kit

PROGRAM

```
#include <lpc214x.h>

void UART0_Init(void)
{
    PLL0CON = 0;
    PLL0FEED=0xAA;
    PLL0FEED=0x55;
    VPBDIV = 1;

    // Fpclk = 12.000.000 MHz
    // DLM,DLH = Fpclk / (19200*16) = 39 = 0x27
    PINSEL0 |= 0x5;          // Select UART0 RXD/TXD
    U0FCR = 0;              // Disable FIFO's
    U0LCR = 0x83;           // 8N1, enable Divisor latch bit
    U0DLL = 0x27;           // baud rate fixed to 19200 @ PCLK = 12 Mhz
    U0DLM = 0;
    U0LCR = 3;              // Disable Divisor latch bit
}

/*-----*/
/* Function to send one char. to Serial Port */
void sout(unsigned char dat1)
{
    while(!(U0LSR & 0x20));    //Wait for Tx Buffer Empty
    U0THR = dat1;             //Send to UART1
}

/*-----*/
int main (void)
{
    int dat;
    UART0_Init();

    do
    {
        if(U0LSR & 1)         /* Check for RDR (Receiver Data Ready) */
        {
            dat = U0RBR;       // Receive Data from Serial Port
            sout(dat);         // Send Data to Serial Port
        }
    }while(1);
}
```

OUTPUT

Data was serially transmitted

RESULT

Thus a C program was written and executed for serial transmission and reception using on-chip UART in LPC2148 kit.